Yang Yan

Ph.D. Candidate, Zhejiang University / Westlake University

AI for Science | Computer Vision | Representation Learning | Structural Biology

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Research Profile

AI-for-science Ph.D. candidate developing foundation models and scalable ML systems for cryo-EM particle analysis and heterogeneous reconstruction. First author of a ***Nature Methods*** study introducing **Cryo-IEF**, the first foundation model for cryo-EM particle processing, pretrained on **65.3M particle images** and transferred to structural classification, pose-aware clustering, particle-quality ranking, and automated reconstruction. Current work develops CryoDECO, a foundation-prior framework for compositional and conformational heterogeneity in cryo-EM. Longer term, my research aims to build structural world models and agentic reconstruction systems that reason over molecular identity, dynamics, and native cellular context.

Education

2023–Present **Ph.D. Candidate in Computer Science** — Zhejiang University / Westlake University
Joint Ph.D. program supervised by Prof. Fajie Yuan and Prof. Huaizong Shen. Research: foundation models for cryo-EM image processing, automated workflows, and heterogeneity analysis.

2020–2023 **M.S. in Translational Medicine (Engineering)** — Xiamen University
National Institute of Diagnostics and Vaccine Development in Infectious Diseases, supervised by Prof. Ningshao Xia. Research: machine learning for medical image analysis.

2016–2020 **B.S. in Electrical Information Engineering** — Northeastern University (Qinhuangdao)
Research included machine-learning applications in localization.

Selected Publications

A comprehensive foundation model for cryo-EM image processing 2026
Yang Yan, Shiqi Fan, Fajie Yuan, Huaizong Shen
Nature Methods, 23(1), 88–95. DOI: 10.1038/s41592-025-02916-8. Introduced Cryo-IEF as the first cryo-EM particle-processing foundation model and established a foundation-model paradigm for automated particle analysis.

Artificial intelligence foundation model automates cryo-EM structure determination 2026
Yang Yan, Huaizong Shen
Nature Methods Research Briefing, 23, 26–27, DOI: 10.1038/s41592-025-02917-7.

CryoDECO: Deconstructing Extreme Compositional and Conformational Heterogeneity in Cryo-EM via Foundation Model Priors 2026
Yang Yan, Yanwanyu Xi, Shiqi Fan, Yifei Wang, Ziyun Tang, Fajie Yuan, Huaizong Shen
LangTaoSha Preprint Server. DOI: 10.65215/LTSpreprints.2025.12.30.000075. SOTA foundation-prior model for heterogeneous cryo-EM reconstruction, advancing high-throughput panoramic structural biology.

Unsupervised cryo-EM images denoising and clustering based on deep convolutional autoencoder and k-means++ 2022
D. Zhang, **Yang Yan**, Y. Huang, B. Liu, Q. Zheng, J. Zhang, N. Xia
IEEE Transactions on Medical Imaging, 42(5), 1509–1521. Deep autoencoder and balanced K-means++ framework for cryo-EM denoising and clustering.

Machine learning for structure determination in single-particle cryo-electron microscopy: A systematic review 2021
J. G. Wu, **Yang Yan**, D. X. Zhang, B. W. Liu, Q. B. Zheng, X. L. Xie, S. Q. Liu, S. X. Ge, et al.
IEEE Transactions on Neural Networks and Learning Systems, 33(2), 452–472. Review of ML methods for single-particle cryo-EM structure determination.

Selected earlier medical-imaging publications 2022–2023
S. Li, B. Liu, S. Li, X. Zhu, **Yang Yan**, D. Zhang; D. Zhang, B. Liu, Y. Huang, **Yang Yan**, et al.
Complex & Intelligent Systems, 2022; Journal of Digital Imaging, 2023. Deep-learning and ordinal-regression methods for bone-age assessment.

Research Highlights

- **First cryo-EM foundation model:** introduced **Cryo-IEF**, the first foundation model developed for cryo-EM particle processing, pretrained on **∼65M particle images** and establishing a new paradigm for automated particle analysis in structural biology.
- **Automated reconstruction system:** developed **CryoWizard**, a fully automated computational pipeline for single-particle cryo-EM reconstruction that streamlines raw data processing, particle ranking, and high-resolution 3D structure determination without manual intervention.
- **Foundation-prior heterogeneous reconstruction:** developed **CryoDECO** to use pretrained particle representations as priors for ab initio heterogeneous reconstruction, targeting compositional and conformational mixtures.
- **Open-source Python package for cryo-EM data processing:** developed `cryodata`, a PyPI-installable data layer that converts CryoSPARC particle jobs into PyTorch-ready datasets with MRC/MRCS preprocessing, FFT/Hartley representations, LMDB storage, balanced sampling, and CryoSPARC-to-RELION metadata conversion.

Research Software and Open Source

2023–2026 **Cryo-IEF: First Foundation Model for Cryo-EM Particle Processing**
https://github.com/westlake-repl/Cryo-IEF
Designed and released the Cryo-IEF ecosystem, pretrained on ∼65M cryo-EM particle images. Built PyTorch inference and downstream tooling for CryoRanker and CryoClustering, enabling practical structural classification, pose clustering, and particle-quality assessment.

2024–2026 **CryoDECO: SOTA Heterogeneous Reconstruction with Foundation Model Priors**
https://github.com/yanyang1998/CryoDECO
Developed a prior-guided heterogeneous reconstruction framework that uses Cryo-IEF representations to reduce random ab initio initialization and disentangle compositional classification from 3D reconstruction.

2024–2025 **CryoWizard: Fully Automated Single-Particle Cryo-EM Reconstruction Pipeline**
https://github.com/SMART-StructBio-AI/CryoWizard
Built and extended an end-to-end computational pipeline integrating CryoRanker with CryoSPARC, streamlining raw movies, micrographs, or particles to high-resolution 3D volumes without manual intervention through command-line, web, and Chrome-extension interfaces.

2023–2026 **cryodata: Open-Source Python Package for Cryo-EM Data Processing**
https://github.com/SMART-StructBio-AI/cryoief-data
Created the **reusable data layer** used by Cryo-IEF, CryoDECO, and CryoWizard to convert raw CryoSPARC particle outputs into PyTorch-ready training and inference datasets. The package supports MRC/MRCS preprocessing, LMDB-backed dataset creation, Fourier/Hartley feature generation, balanced sampling, PyTorch data loading, and CryoSPARC-to-RELION metadata conversion. Packaged and documented the toolkit for `pip install cryodata`, turning lab-specific preprocessing into reproducible open-source scientific ML infrastructure.

Technical Skills

ML / Vision	Self-supervised learning, representation learning, foundation models, clustering, autoencoders, generative/reconstruction models, image denoising.
Structural Biology	Single-particle cryo-EM, cryo-ET, CryoSPARC/RELION workflows, ChimeraX visualization, MRC/MRCS data formats.
Systems	Multiprocessing data pipelines, parallel GPU training/inference.

Research Agenda

I aim to build foundation-model priors and agentic reconstruction systems for cryo-EM/cryo-ET that infer molecular identity, conformational dynamics, and native cellular context from large-scale imaging data. My near-term focus is representation-guided heterogeneous reconstruction; my long-term goal is AI-guided panoramic structural biology for complex molecular mixtures.